

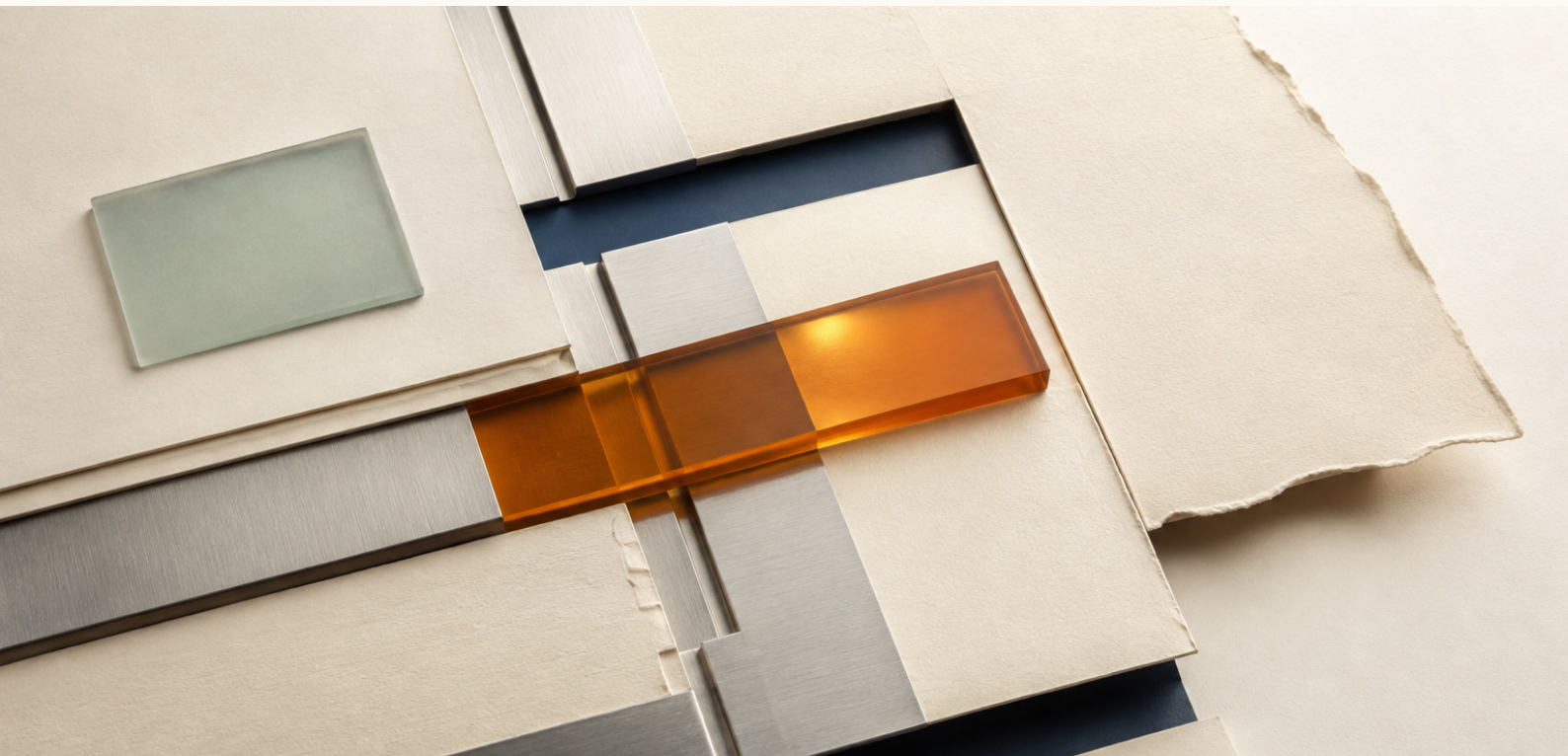
PAYMENTS / TREASURY / DIGITAL ASSETS

Accounting for an Omnichain Dollar

USDT0 token-unit backing, assurance perimeter, and bank-control implications.

PROTOCOL-BACKING CONCLUSION

Documented direct USDT0 supply reconciles to observed USDT backing at 1.000298x. This is a token-unit result: par within measurement tolerance, not an economic reserve cushion or issuer-level reserve opinion.



TSOLMONDORJ NATSAGDORJ / SUWAPPU RESEARCH

Working paper revised 6 August 2026. Report published 6 August 2026.

SUWAPPU.BOT/RESEARCH

BANKING TAKEAWAY

Protocol backing reconciles to par. Issuer assurance is a separate question.

At 01:53 UTC on 1 August 2026, the verified Ethereum lockbox held 3.453608822B USDT against 3.452579540B USDT0 across the complete documented direct-supply perimeter. Coverage was 1.000298x. The 1.029283M-unit difference - about three basis points - is classified as par within measurement tolerance, not as an economic reserve cushion.

3.4536B

USDT IN BACKING ACCOUNT

3.4526B

DIRECT USDT0 SUPPLY

1.000298x

TOKEN-UNIT COVERAGE

WHAT THIS CONTROL MEASURES

Public state can read USDT in the canonical backing account, aggregate totalSupply() across every documented direct USDT0 leg, version the accounting perimeter and alert on a token-unit mismatch. The head check covers 20 direct supply legs in one session.

WHAT IT CANNOT ESTABLISH

It does not test Tether's reserve portfolio, a holder's legal claim, USDT redemption capacity, stressed liquidity, technical versus legal finality, net messages in flight, or whether the deployment registry is exhaustive. Those require separate evidence.

CONTROL CONCLUSION

Use the token-unit reconciliation as a repeatable first-line monitoring control. Do not promote it into a credit, liquidity, settlement-finality or reserve conclusion without the issuer, legal and operational evidence required for those decisions.

CONTROL FINDING

The two corrections are the control finding.

The first error created false excess coverage; the second created a false protocol-backing shortfall. Both came from population and reference-data failures that a bank control framework should catch before a ratio reaches a risk committee.

VERSION	REPORTED SIGNAL	INTERPRETATION	CONTROL FAILURE
V1	1.042 endpoint	False surplus	Direct-supply population truncated
V2	0.513-0.588	False shortfall	Wrong Polygon backing address
V3	1.000298 head	Par / ~3bp	Complete documented direct-supply population

CONTROL IMPLICATIONS

- 01

Population governance

V1 omitted 134.8M USDT0 units. Completing the documented population removed 96% of the reported arithmetic difference. Scope needs an owner, effective date, inclusion rule and completeness check.
- 02

Canonical-account governance

V2's control address read 0.02 USDT while the canonical Polygon predicate held 1.22B-1.39B USDT. That is a reference-data failure, not evidence of backing deterioration.
- 03

Timestamp discipline

Cross-chain balances need a common observation time. The panel resolves each chain to the target timestamp; the migration window is rescanned every six hours so settlement flow is not misread as a backing mismatch.

OPERATING RULE	SR 26-2 excludes simple arithmetic and deterministic rules from its model definition. Govern this ratio primarily as a controlled population, reference-data and use-of-output process.
----------------	---

OBSERVED PERIMETER

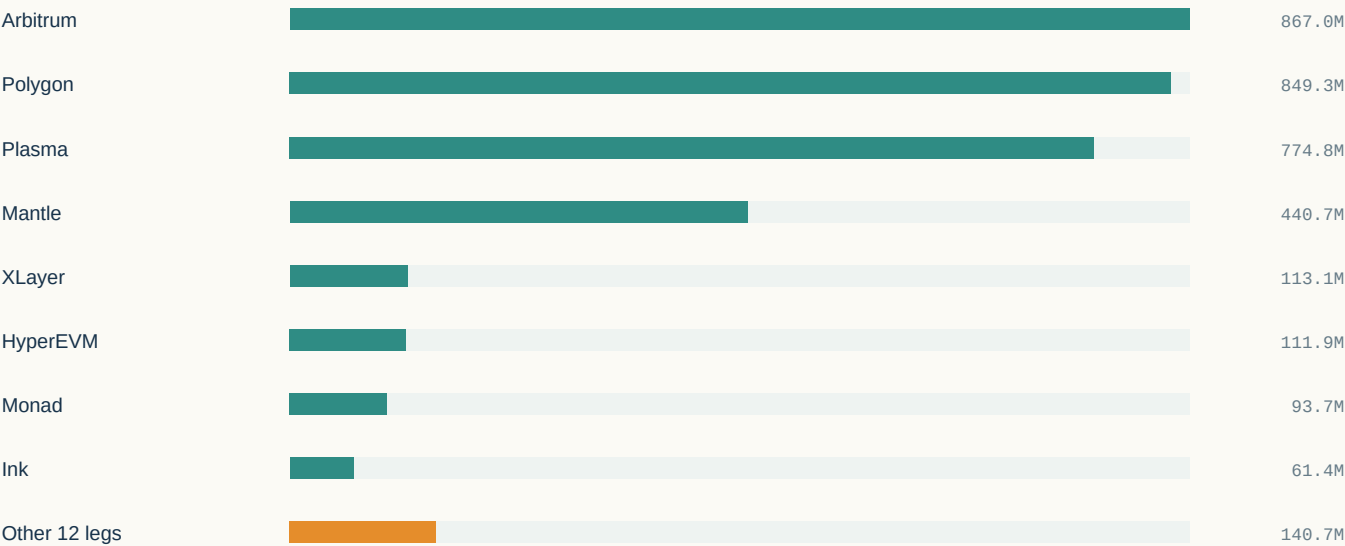
The measured perimeter contains twenty direct supply legs.

The observable identity compares USDT token units in the verified Ethereum backing account with aggregate USDT0 totalSupply() across the documented direct deployments. HyperCore is contained in HyperEVM and is not double-counted; Tron and TON sit outside this direct-supply perimeter as Legacy Mesh.

1 AUG 2026 / 01:53 UTC

3.4536B USDT / 3.4526B USDT0 = 1.000298x

DIRECT SUPPLY PERIMETER



20 direct legs measured

0 documented legs excluded

HyperCore contained

SOURCE / HEAD_SNAPSHOT_20260801.JSON

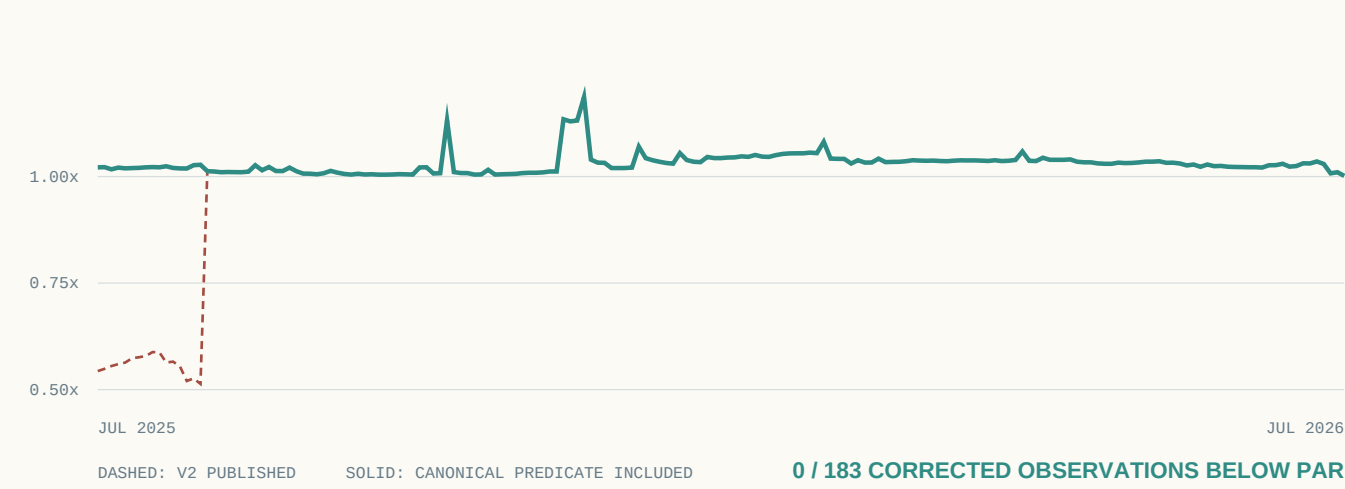
INTERPRETATION

For a bank control, 1.000298x is reconciled to par within measurement tolerance. It is not a USD mark-to-market ratio and says nothing about the composition or adequacy of Tether's underlying reserves. The head reads span roughly one minute rather than one aligned block.

WHY THE SHORTFALL DISAPPEARED

A 1.3B-unit apparent backing shortfall was address-mapping risk.

Before 27 August 2025, Polygon PoS USDT backing sat in the canonical Polygon predicate rather than the USDT0 lockbox. V2 checked a different address and converted an address-map failure into an apparent 41%-49% token-unit shortfall. For a bank monitor, wrong reference data can be as consequential as a wrong balance.



SIX-HOUR MIGRATION BRACKET / 27 AUG 2025

-1,358.8M	+1,258.6M	-98.1M	2.1M
USDT / PREDICATE	USDT / LOCKBOX	USDT0 / ARBITRUM	RESIDUAL UNITS

CONTROL VALIDATION

The 1,358.8M-USDT predicate outflow matched Polygon supply at bracket open (1,358.759M units) to within about 82K units - 0.006% of the flow. USDT0 also announced the backing migration that day. Contemporaneous documentation and chain state describe the same perimeter change.

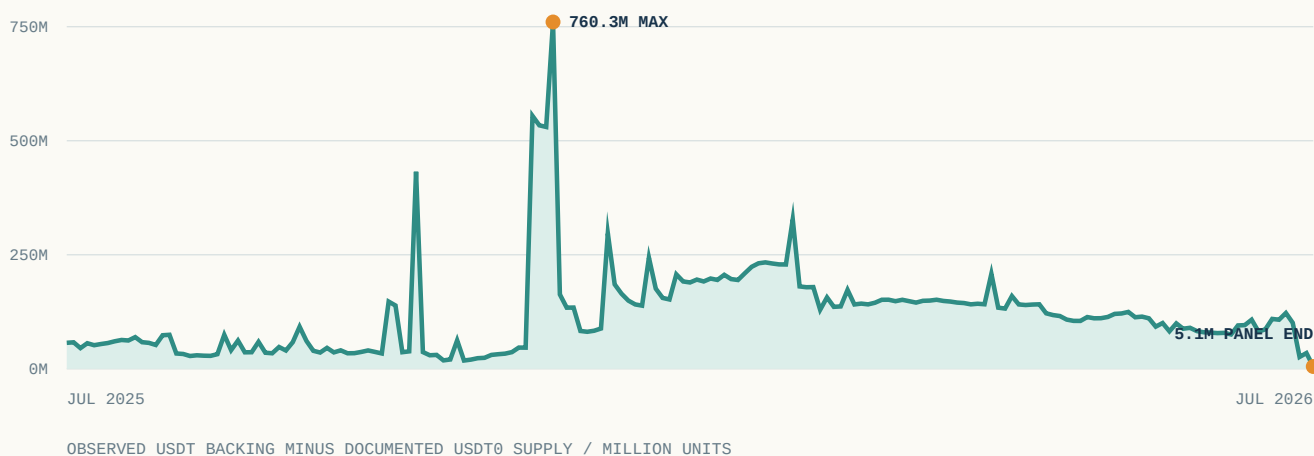
LESSON

Treat contract and address changes as governed reference-data events. Otherwise an accounting-boundary migration can look indistinguishable from a protocol-backing impairment.

HISTORICAL CONTEXT

The backing difference compressed toward par; it was not structural.

Historical corrected token-unit coverage stayed above par at the sampled points, but the arithmetic difference was not stable: it ranged from roughly 15bp to 18.7% of measured supply. By sample end it had compressed to measurement tolerance. The historical maximum should not be carried forward as a risk cushion.

**760.3M**

MAX / 15 DEC 2025

296.9M

31 DEC 2025

122.4M

17 JUL 2026

5.1M

25 JUL / PANEL END

FINAL EIGHT DAYS

-192.9M USDT backing vs -75.7M USDT0 supply

Backing declined by about 117M units more than measured supply. The panel difference fell to 5.1M units; the complete documented head check six days later measured 1.029M units. The data establish the change, not its cause.

INTERPRETATION

Historical excess token-unit coverage should not be carried forward as a current economic buffer. The decision-relevant observation is the endpoint: protocol backing is effectively at par, while issuer, legal, liquidity and message-state risks remain outside this ratio.

PROTOCOL CHECK VS BANK DILIGENCE

A balance ratio is one layer of settlement-asset diligence.

CPMI-IOSCO's stablecoin guidance separates settlement finality, legal claims, credit risk, liquidity risk and stressed convertibility. Public chain state can support a repeatable protocol-backing control; it does not answer those broader settlement-asset questions.

MEASURED FROM PUBLIC STATE

REQUIRES SEPARATE EVIDENCE

- Protocol backing balance**
USDT balanceOf() on the verified Ethereum OAdapter backing account.
 - Documented direct supply**
totalSupply() on each documented direct USDT0 deployment, with decimals and live contracts checked.
 - Versioned accounting perimeter**
Twenty direct supply legs, plus HyperCore containment; Tron and TON classified separately as Legacy Mesh.
- USDT issuer / holder claim**
The balance does not establish Tether reserve quality, holder rights, legal availability, seniority or recoverability.
 - Redemption / stressed liquidity**
The ratio does not test USDT redemption eligibility, market liquidity, convertibility at par or performance under stress.
 - Finality / message / registry state**
Point-in-time stocks do not prove legal finality, net in-flight state, security configuration or registry completeness.

MINIMUM EVIDENCE STACK FOR A BANK

- 01 Token-unit backing reconciliation

02 Controlled population + account map

03 Message state + settlement finality

04 Issuer reserve + assurance evidence

05 Legal claim + redemption terms

06 Stress liquidity + prudential review

THRESHOLD

At ~3bp, any unobserved net item above 1.029M token units can change the sign of the protocol arithmetic. That is a materiality threshold for reconciliation exceptions, not an estimate of issuer, legal or liquidity risk.

SUGGESTED OPERATING MODEL

Automate the reconciliation; govern the evidence it cannot see.

The practical architecture separates deterministic token-unit arithmetic from population governance, cross-chain operational state and issuer-level credit, legal and liquidity evidence. Different controls answer different questions; none should be promoted into the others.

CONTROL LAYERS

01

Daily token-unit reconciliation

Read canonical USDT backing and every documented direct USDT0 supply leg; version the timestamp and population; alert on a defined token-unit exception.

02

Reference-data change control

Detect deployments, migrations, upgrades, containment changes and registry revisions. Reverify contract identity, decimals, account mapping and effective date before inclusion.

03

Cross-chain operational state

Monitor net messages in flight, execution queues, permissions and settlement finality separately from point-in-time token stocks.

04

Issuer / legal / liquidity overlay

Combine Tether reserve and assurance evidence with legal claims, redemption access, normal/stressed liquidity and the bank's own prudential treatment.

REPRODUCIBILITY TRAIL

`data/usdt0_timeseries.csv`

183-row panel + retained wrong-address control

`data/polygon_predicate_prebreak.json`

canonical Polygon backing backfill

`data/head_snapshot_20260801.json`

complete documented head measurement

`data/buffer_dynamics.json`

flow coupling + terminal difference

`code/collect_usdt0.py`

panel collection harness

`code/head_snapshot.py`

current-universe snapshot

OPEN METHODOLOGY AND SOURCE DATA

suwappu.bot/research/replication

CONCLUSION

The protocol is at par on observed token units. No broader reserve conclusion follows.

The current documented USDT0 perimeter reconciles to observed USDT backing at 1.000298x. The durable control insight is that the population and account map can be governed and replayed. Tether reserve quality, redemption, legal claims, stressed liquidity and settlement finality remain separate diligence questions.

01

For treasury and payments

Use the public-state check as a repeatable monitoring overlay, not as issuer assurance. It can flag protocol divergence or stale reference data; it cannot establish suitability as a settlement asset.

02

For risk and control governance

Version the registry, account map, method and exceptions. SR 26-2 excludes simple deterministic arithmetic from its model definition; govern the reconciliation as a controlled data and use-of-output process, subject to internal classification.

03

For issuers and infrastructure partners

Publish a dated machine-readable registry of each direct supply leg, backing account, containment relationship and migration. Pair it with evidence for message state, legal rights and operational change control.

DISCLOSURES

This report is research, not a reserve attestation, audit or legal opinion, credit rating, PFMI assessment, regulatory classification, prudential-capital opinion or investment recommendation. Suwappu builds cross-chain execution infrastructure and holds operational USDT and USDT0 balances incidental to running it; no directional position informed this analysis. Tether, Everdawn Labs and other named parties did not review the work before publication. The working paper is canonical where this report and the paper differ.

SELECTED SOURCES

USDT0: Developer Guide, Deployments, Legacy Mesh; Polygon migration notice (2025-2026)

LayerZero: Omnichain Fungible Token (OFT) Standard (accessed 6 Aug 2026)

Tether: Legal Terms and Transparency (accessed 6 Aug 2026; issuer-authored evidence)

CPMI-IOSCO: Application of the PFMI to stablecoin arrangements (2022)

Federal Reserve / FDIC / OCC: Supervisory Guidance on Model Risk Management, SR 26-2 (2026)

Basel Committee: Basel Framework, SCO60 - Cryptoasset exposures (in force 1 Jan 2026)

WORKING PAPER + EVIDENCE BUNDLE

SUWAPPU.BOT/RESEARCH/REPLICATION